

Operator Discovery Report: restart_controller_adaptive_escape_bottleneck

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic restart controller that triggers additional restarts after short stagnation, with a larger perturbation-driven seed pool to escape two-cluster bottlenecks and grid-like traps while keeping restart growth ca
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:3:7:2.
- Surviving candidate: False.
- Problem class where it helps: two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 08bac8fd1984f5ef.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Runtime inflation is high at 2.055607.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 344.039829, distance-eval delta 8604.0.
- cheapest_insertion_2opt gap delta -0.00408, runtime delta 321.847243, distance-eval delta 7860.571429.
- random_restart_2opt gap delta 0.0, runtime delta -171.714286, distance-eval delta -9090.285714.
- sparse_three_opt gap delta 0.031858, runtime delta 83.963757, distance-eval delta 4854.142857.
- clustered_local_search gap delta -0.000591, runtime delta -16.3867, distance-eval delta -412.0.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.
- clustered_tsp operator gap 0.110431 vs baseline 0.110853.

- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.003363.
- Runtime inflation: 2.055607.
- Distance-evaluation inflation: 1.878498.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.003363.
- fixed_restart_budget mean gap 0.117954 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.117688 and delta vs full operator -0.000266.